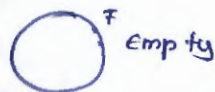
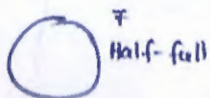
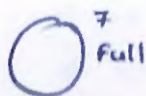


①



x_{ij} - # of bottle bottles of type i given to j^{th} individual

$i \geq 0$ and Integers

$j = 1, 2, 3$

Bottle type		Individual			
		1	2	3	
	full	x_{11}	x_{12}	x_{13}	7
	half full	x_{21}	x_{22}	x_{23}	7
	empty	x_{31}	x_{32}	x_{33}	7
		7	7	7	

s.t.

Individual receives ;

$$x_{11} + x_{21} + x_{31} = 7$$

$$x_{12} + x_{22} + x_{32} = 7$$

$$x_{13} + x_{23} + x_{33} = 7$$

Bottle type total ;

$$x_{11} + x_{12} + x_{13} = 7$$

$$x_{21} + x_{22} + x_{23} = 7$$

$$x_{31} + x_{32} + x_{33} = 7$$

If the total amount / full amount of a bottle is given by a ;

$$\text{Total amount in all bottles} = 7 \times a + 7 \times \frac{a}{2} + 7 \times 0$$

$$= \frac{7a + \frac{7a}{2}}{2} = 10.5a$$

Since each individual must
 receive the same
 quantity
 one will receive

$$= \frac{10.5a}{3} = 3.5a$$

Individuals getting the same quantity:

$$ax_{11} + \frac{a}{2}x_{21} + 0x_{31} = 3.5a$$

$$x_{11} + \frac{1}{2}x_{21} = 3.5$$

$$ax_{12} + \frac{a}{2}x_{22} + 0x_{32} = 3.5a$$

$$x_{12} + \frac{1}{2}x_{22} = 3.5$$

$$ax_{13} + \frac{a}{2}x_{23} + 0x_{33} = 3.5a$$

$$x_{13} + \frac{1}{2}x_{23} = 3.5$$

$$\text{Max } Z = 0x_{11} + 0x_{12} + \dots + 0x_{23}$$

(Since all the bottles are equally distributed among individuals, there is no any optimization. This includes only the feasibility)

Again, this's not an Optimization problem.

x_1 - # of camels to Tarek

x_2 - # of camels to Sharif

x_3 - # of camels to Malsa

x_4 - # of camels to Charity.

$x_1, x_2, x_3, x_4 \geq 0$ and integers.

$$\text{Max } Z = 0x_1 + 0x_2 + 0x_3 + 0x_4$$

$$x_4 = 1$$

Since total # of camels in the herd is an odd number

$$x_1 + x_2 + x_3 + x_4 = \text{an odd number}$$

$$x_1 + x_2 + x_3 + 1 = \text{ " " " }$$

$$\underbrace{\hspace{10em}}_{2y}$$

$$\therefore x_1 + x_2 + x_3 = 2y = \text{an even number.}$$

where y is a positive integer.

Tarek: $x_1 \geq \frac{1}{2}(x_1 + x_2 + x_3 + 1)$

$$\frac{1}{2}x_1 - \frac{1}{2}x_2 - \frac{1}{2}x_3 - \frac{1}{2} \geq 0$$

$$x_1 + x_2 - x_3 - 1 \geq 0$$

$$x_1 - x_2 - x_3 \geq 1$$

Sharif: $x_2 \geq \frac{1}{3}(x_1 + x_2 + x_3 + 1)$

$$\frac{2}{3}x_2 - \frac{1}{3}x_1 - \frac{1}{3}x_3 - \frac{1}{3} \geq 0$$

$$2x_2 - x_1 - x_3 \geq 1$$

Malso;

$$x_3 \geq \frac{1}{7} (x_1 + x_2 + x_3 + 1)$$

$$\frac{6}{7} x_3 - \frac{1}{7} x_1 - \frac{1}{7} x_2 \geq \frac{1}{7}$$

$$6x_3 - x_1 - x_2 \geq 1$$

③

x_i - the selection of route i

$$x_i = \begin{cases} 0 & , \text{ if the } i^{\text{th}} \text{ route is not selected} \\ 1 & , \text{ if the } i^{\text{th}} \text{ route is selected.} \end{cases}$$

R_i - Route i

If R_1 is selected, total distance:

$$ABC \rightarrow 3 \rightarrow 2 \rightarrow ABC$$

$$0 + 16 + 14 + 12 = 42$$

If R_2 is selected, total distance:

$$ABC \rightarrow 5 \rightarrow 3 \rightarrow 4 \rightarrow ABC$$

$$8 + 18 + 15 + 9 = 50$$

If R_3 is selected, total distance: = 66

If R_4 " " " " = 52

If R_5 " " " " = 60

If R_6 " " " " = 54

Minimize Total distance

$$(Z) = 42x_1 + 60x_2 + 66x_3 + 52x_4 + 60x_5 + 84x_6$$

Customer 01;

$$x_3 + x_5 + x_6 \geq 1$$

Customer 02;

$$x_1 + x_3 + x_4 + x_5 \geq 1$$

Customer 03;

$$x_1 + x_2 + x_3 + x_4 + x_6 \geq 1$$

Customer 04;

$$x_2 + x_5 \geq 1$$

Customer 05;

$$x_2 + x_3 + x_4 + x_6 \geq 1$$

$$x_1, x_2, x_3, x_4, x_5, x_6 = (0, 1)$$

04

Objective function $\rightarrow$ To maximize the profit.

(2)

x_1 - # of units from product 1
 x_2 - # of " " " " 2
 x_3 - " " " " " 3
 x_4 - " " " " " 4
 x_5 - " " " " " 5
 x_6 - " " " " " 6
 x_7 - " " " " " 7

$Y_1 = \begin{cases} 1, & \text{if product 7 is produced} \\ 0, & \text{if product 7 is not produced} \end{cases}$

$x_2 \begin{cases} x_{21} - \# \text{ of products from product 2} \leq 100 \\ x_{22} - \# \text{ of products from product 2, in addition to 100 initial} \end{cases}$

$$\text{Max } Z = 10x_1 + 22(x_{21} + x_{22}) + 35x_3 + 19x_4 + 55x_5 + 10x_6 + 115x_7 - 2000Y_1$$

s.t.o;

$$x_1 + 2x_{21} + 3x_{22} + 3.7x_3 + 2.4x_4 + 4.5x_5 + 0.7x_6 + 9.5x_7 \leq 720$$

$$x_1 + x_3 \geq 80Y_2$$

$$x_2 + x_4 = 85(1 - Y_2)$$

or,

$$\begin{aligned} x_1 + x_3 &= 80 + MY_2 \\ x_2 + x_4 &= 85 + M(1 - Y_2) \end{aligned}$$

$Y_2 = \begin{cases} 1, & \text{if market condition is satisfied} \\ 0, & \text{if market condition is not satisfied.} \end{cases}$

$Y_2 = \begin{cases} 0, & \text{market condition is satisfied} \\ 1, & \text{o/w.} \end{cases}$

$$x_1, x_{21}, x_{22}, x_3, x_4, x_5, x_6, x_7 \geq 0 \text{ and integers.}$$